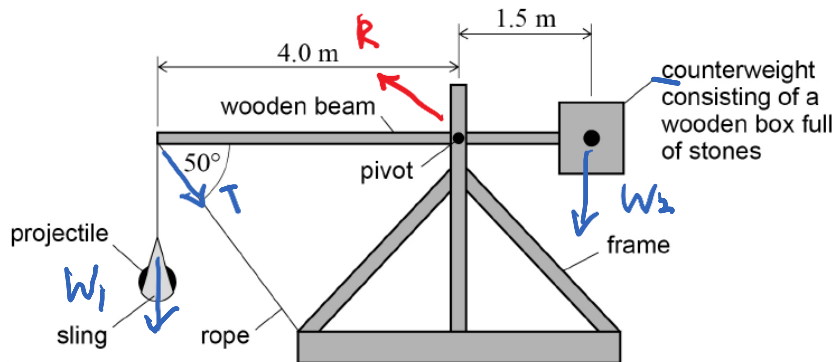


- 1 (a) The centre of mass/gravity of the beam and box is at the pivot B1
 For equilibrium, moments balance / sum of the moments is zero at this position B1

- (b) Take moments about pivot, C1
 Clockwise moment = Anticlockwise moment C1
 $610 \times 9.81 \times 1.5 = 250 \times 4 + T \sin 50^\circ \times 4.0$ C1
 $T = 2600 \text{ (N)}$ A1

- (c) 1. Draw direction of force R as shown.



B1

2 Assume that the weight of the beam and empty box is negligible